

Question 1

a.

$$\text{Max utilization} = (m/b)/\text{RTT} = (8000/10\,000\,000)/0.05\text{s} = 0.016$$

$$\text{Max data rate} = \text{network capacity} * \text{max utilization} = 10\text{ Mbps} * 0.016 = 0.16\text{ Mbps}$$

b.

Sending window = 10:

$$\begin{aligned}\text{Max data rate} &= (\text{sending window size} / \text{frame size}) / \text{RTT} \\ &= 10 * 8000 / 0.05 = 1.6\text{ Mbps}\end{aligned}$$

Sending window = 50:

$$\begin{aligned}\text{Max data rate} &= (\text{sending window size} / \text{frame size}) / \text{RTT} \\ &= 50 * 8000 / 0.05 = 8.0\text{ Mbps}\end{aligned}$$

c.

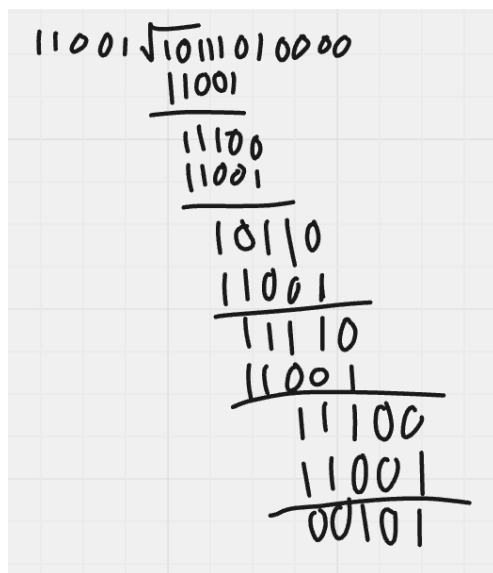
$$\begin{aligned}\text{Sending window size} &\geq \text{data rate} * \text{RTT} / \text{frame size} \\ &\geq 10\,000\,000\text{ bps} * 0.05\text{s} / 8000 \\ &\geq 62.5 \\ &\geq 63\end{aligned}$$

The sending window size would need to be at least 63.

d.

The above sending window assumes that there is no loss at all, and all frames and ACKs are received properly. If this isn't the case, there will still be a need to wait for ACKs and we will lose utilization. For a situation like this, a bigger maximum sending window size would let us begin to send more frames before requiring a successful ACK.

Question 2



Full checksum = 10111010101.

Question 3

a.

$$\begin{aligned}\text{Propagation delay} &= \text{distance} / \text{signal propagation speed} \\ &= 90\text{m} / 3 \times 10^8\text{m/s} \\ &= 3.0 \times 10^{-7}\text{s}\end{aligned}$$

$$\begin{aligned}\text{Transmission time} &= \text{frame size} / \text{transmission speed} \\ &= 1500 \text{ bytes} / 50 \times 10^6 \text{ bytes/s} \\ &= 3.0 \times 10^{-5}\text{s}\end{aligned}$$

$$\text{Ratio} = \tau / t = 3.0 \times 10^{-7}\text{s} / 3.0 \times 10^{-5}\text{s} = 0.01$$

Due to the low τ / t ratio, CSMA would be preferable.

b.

$$\begin{aligned}\text{Propagation delay} &= \text{distance} / \text{signal propagation speed} \\ &= 600\text{m} / 3 \times 10^8\text{m/s} \\ &= 2.0 \times 10^{-6}\text{s}\end{aligned}$$

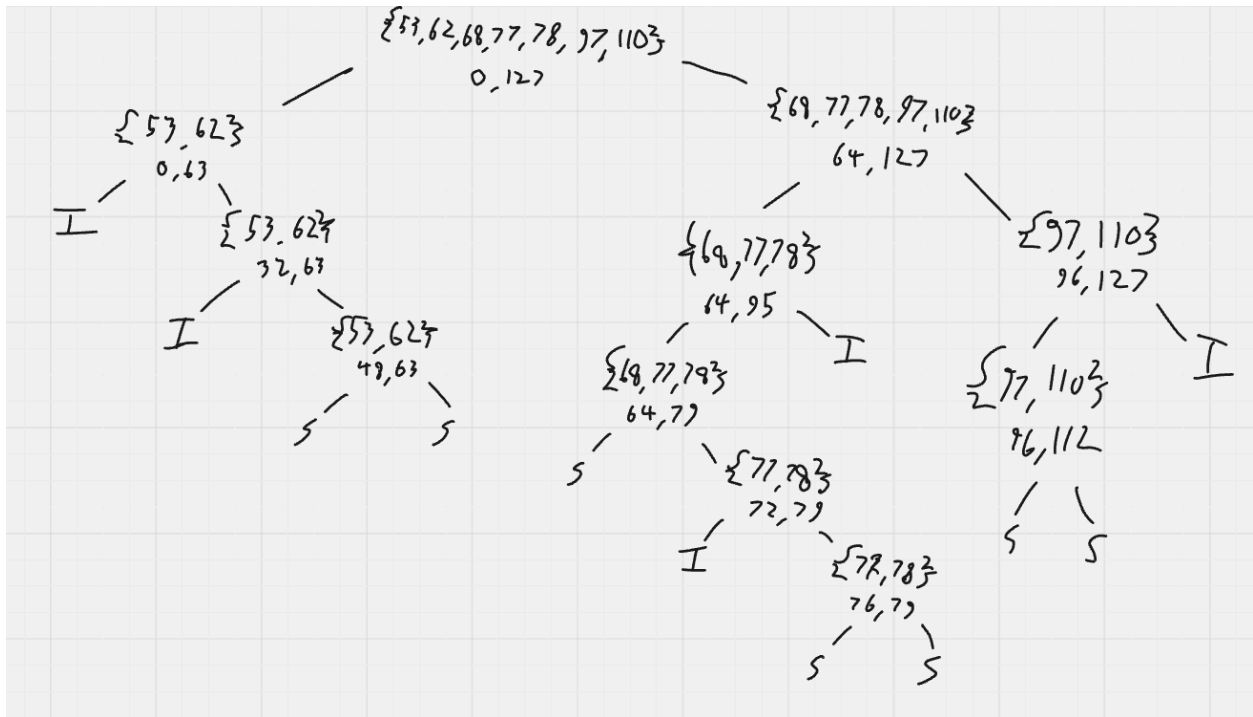
$$\begin{aligned}\text{Transmission time} &= \text{frame size} / \text{transmission speed} \\ &= 1000 \text{ bytes} / 1.25 \times 10^9 \text{ bytes/s} \\ &= 8.0 \times 10^{-7}\text{s}\end{aligned}$$

$$\text{Ratio} = \tau / t = 2.0 \times 10^{-6}\text{s} / 8.0 \times 10^{-7}\text{s} = 2.5$$

At high τ / t ratios, CSMA becomes much worse, so ALOHA would be preferable.

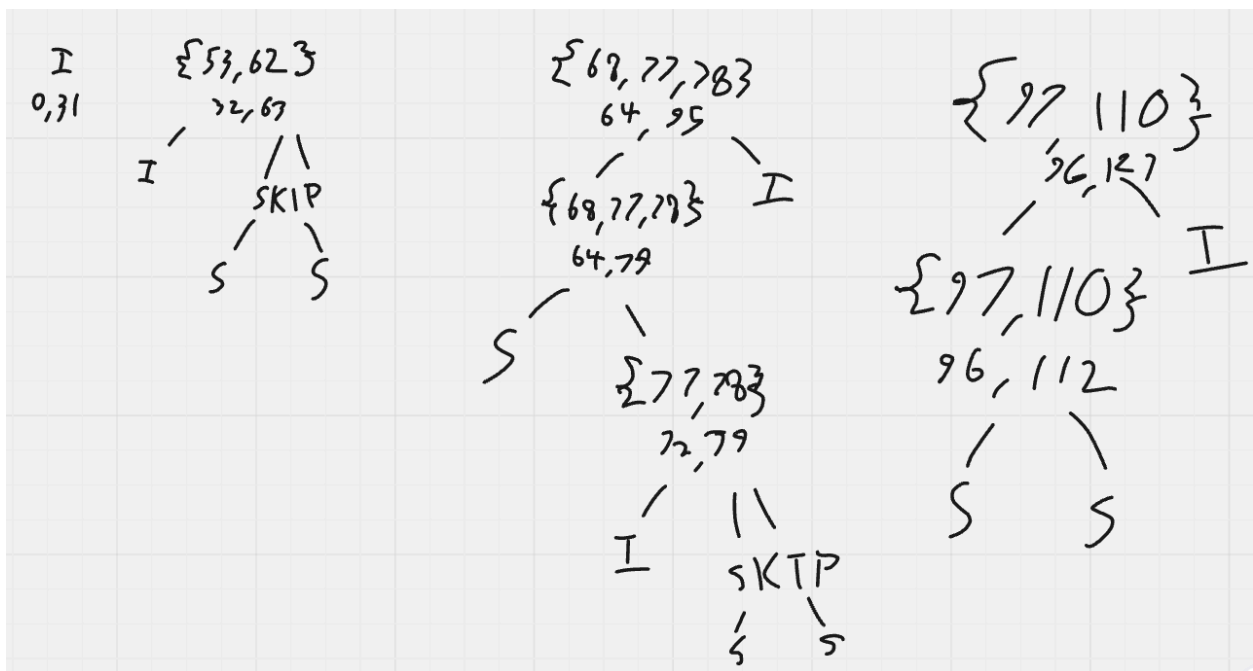
Question 4

a.



Traversal time = slots * 100 μ s = 23 * 100 μ s = 2300 μ s

b.



Traversal time = slots * 100 μ s = 18 * 100 μ s = 1800 μ s